

Task 1:

Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Answer:

While refreshing the Flipkart homepage, I captured the network traffic using Wireshark. I identified packets from different OSI layers by examining their headers.

Application Layer: I identified a DNS/HTTPS packet. The header contains application-related information such as DNS query details or TLS information.

Transport Layer: I identified a TCP packet. Its header contains source port, destination port, sequence number, and acknowledgment number.

Internet Layer: I identified an IPv4/IPv6 packet. Its header contains source IP address, destination IP address, and TTL/Hop Limit.

Network Access Layer: I identified an Ethernet II packet. Its header contains source MAC address, destination MAC address, and EtherType.

Thus, Wireshark helped me observe how data from the Flipkart website is encapsulated and transmitted through different layers of the OSI model.

Task 2:

Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

Answer:

OSI Model Layer	TCP/IP Model Layer	Real-World Example
7. Application	Application	WhatsApp message: You type and send "Hello".
6. Presentation	Application	WhatsApp encrypts the message before sending it.
5. Session	Application	WhatsApp maintains the communication session between sender and receiver.
4. Transport	Transport	TCP/UDP handles data delivery between your phone and the server.
3. Network	Internet	IP addresses are used to route the WhatsApp data to the destination.
2. Data Link	Network Access	Wi-Fi/Ethernet uses MAC addresses to transfer data within the local network.
1. Physical	Network Access	Wi-Fi radio signals or Ethernet cables physically carry the data.

Task 3:

Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Answer:

Source Port: Identifies the port number of the application sending the data.

Destination Port: Identifies the port number of the application receiving the data.

Sequence Number: Helps TCP arrange the received data in the correct order.

Acknowledgment Number: Indicates the next byte of data that the receiver expects to receive.

Header Length: Shows the size of the TCP header.

Flags: Control the TCP connection. Common flags are SYN, ACK, FIN, RST, PSH and URG.

Window Size: Tells the sender how much data the receiver can accept at a time.

Checksum: Checks whether the TCP segment has been corrupted during transmission.

Urgent Pointer: Indicates the position of urgent data when the URG flag is used.

Options: Contains additional TCP information, such as Maximum Segment Size (MSS) and window scaling.

Task 4:

Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

Answer:

Message Sent from Device to Instagram Server – TCP/IP Layers

Step 1 – Application Layer:

Your Device → Instagram

What happens: You type and send a message on Instagram. The application prepares the message to be sent over the network.

Step 2 – Transport Layer:

Message → Segments

What happens: The message is divided into smaller pieces called segments. TCP adds information such as source port, destination port, sequence number, and acknowledgment information.

Step 3 – Internet Layer:

IP Addresses Added

What happens: The device adds the source IP address and destination IP address. These addresses help routers determine where the data should go.

Step 4 – Network Access Layer:

Data → Wi-Fi/Ethernet → Internet

What happens: The data is converted into frames and sent through Wi-Fi or Ethernet. MAC addresses are used to deliver the data to the next device on the network.

Step 5 – Instagram Server:

Data Received → Message Reconstructed

What happens: Instagram's server receives the data, processes the packets/segments, and reconstructs the original message so the application can handle it.

Task 5:

Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

ANSWER:

Packet Selected: TCP packet

TCP/IP Layer: Transport Layer

How I identified it:

I identified the packet as a TCP packet by checking the Protocol column in Wireshark, where it was shown as TCP. In the packet details, I could also see TCP header fields such as Source Port, Destination Port, Sequence Number, Acknowledgment Number, and Flags.

Real-world action:

This packet represents data being transferred when I perform an action in an app, such as sending a WhatsApp message. TCP helps deliver the data reliably and keeps the segments in the correct order.

Conclusion:

The selected packet belongs to the Transport Layer of the TCP/IP model and represents the reliable transfer of application data between my device and the server.